Software Test Plan (STP) - Smart Parking Management System

Project: Smart Parking Management System (SPMS)
Version: 1.0
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1. Introduction

Purpose: This document defines the test plan for the Smart Parking Management System (SPMS) v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers SPMS features such as registration/login, lot and slot management, real-time availability, slot reservation, automatic slot allocation, vehicle entry/exit, fee calculation and payments, reports and notifications. Physical gate hardware (barriers, ANPR cameras, sensors) and payment gateway internals are excluded.

References: SPMS SRS v1.0, SPMS SAD (Design Specification) v1.0, OWASP ASVS 4.0, WCAG 2.1.

Definitions: SPMS (Smart Parking Management System), JWT (JSON Web Token), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing).

2. Test Items

- Authentication & user module
- Lot, slot & tariff management module
- Real-time availability module
- Reservation module
- Slot allocation engine
- Vehicle entry/exit (gate) module
- Billing & payment module
- Reports, history & notification module
- Admin dashboard / attendant console UI

3. Features to be Tested

Features mapped to SRS requirement IDs:
- SPMS-F-001: Driver registration

- SPMS-F-002: Login & JWT issue
- SPMS-F-003: Role-based access control
- SPMS-F-004: Account lockout after 5 failures
- SPMS-F-010: Manage lots / levels / slots
- SPMS-F-011: Configure tariffs
- SPMS-F-012: Slot maintenance status
- SPMS-F-020: Display free-slot counts
- SPMS-F-021: Live status push ≤ 5 s
- SPMS-F-030: Reserve a slot
- SPMS-F-031: Prevent double booking
- SPMS-F-032: Modify / cancel with refund
- SPMS-F-033: Auto-release no-shows
- SPMS-F-040: Auto-allocate suitable slot
- SPMS-F-041: Attendant allocation override
- SPMS-F-050: Record vehicle entry
- SPMS-F-051: Validate reservation QR
- SPMS-F-052: Reject duplicate entry
- SPMS-F-053: Record vehicle exit
- SPMS-F-060: Fee calculation
- SPMS-F-061: Online & cash payment
- SPMS-F-062: Digital receipt
- SPMS-F-063: Block unpaid exit / waiver
- SPMS-F-070: Occupancy & revenue dashboard
- SPMS-F-071: CSV export
- SPMS-F-072: Email notifications
- SPMS-F-073: Driver history & receipts
- SPMS-NF-001: API response time
- SPMS-NF-002: Gate transaction time
- SPMS-NF-003: Availability & backup
- SPMS-NF-004: Usability & accessibility
- SPMS-NF-005: Scalability
- SPMS-NF-006: Maintainability

4. Features Not to be Tested

- Payment gateway internal processing and settlement (vendor responsibility; tested only in sandbox mode)
- Physical hardware – boom barriers, ANPR cameras, slot sensors (out of scope, simulated via API)
- Email delivery beyond the SMTP server (spam filtering / inbox placement)

5. Test Approach / Strategy

Levels:
- Unit tests (module-level, pytest)
- Integration tests (API ↔ PostgreSQL/Redis, API ↔ Payment Gateway sandbox)

- System tests (end-to-end SPMS functionality through the UI, Playwright)
- Acceptance tests (UAT)

Types:
- Functional testing (core features, boundary values: grace period, daily cap, reservation window)
- Concurrency testing (parallel bookings for the last slot – no double booking)
- Regression testing (CI on every pull request)
- Performance testing (response time, load with 200 concurrent users)
- Usability testing (UI clarity, accessibility)

Entry Criteria: Stable build delivered, test data available, test environment ready.
Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

5.1 Security Validation

- Validate password handling (bcrypt hashing, not logged) and account lockout
- TLS 1.2+ verification
- JWT expiry/tampering and role-based access (RBAC) checks
- Payment webhook signature verification; no card data stored
- Fuzzing for input fields (plate number, login, reservation forms) – SQL injection / XSS (OWASP ZAP)
- Penetration testing of authentication flows and rate limiting

6. Test Environment

Hardware: Developer laptops, one Docker host / cloud VM, Android phone and tablet (gate console), optional QR scanner.
Software: SPMS app v1.0 (React + FastAPI), PostgreSQL 15, Redis 7, payment gateway sandbox (test mode), SMTP sandbox.
Tools: Playwright/Selenium (UI automation), Postman (API), pytest (unit/integration), Locust/JMeter (performance), OWASP ZAP (security), GitHub Issues/Jira (defect tracking).
Test Data: Dummy driver accounts, vehicles, parking lots/slots, tariffs, reservations and parking sessions.

7. Test Schedule

Milestones:
- Test case design: 20-Oct-2026
- Environment setup: 24-Oct-2026
- Test execution start: 02-Nov-2026
- Test execution end: 13-Nov-2026
- UAT: 16-Nov-2026 to 18-Nov-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Rahul Prasanna	Prepare plan, coordinate execution
Test Engineer	Rufus Thomas	Design & execute test cases, log defects
Developer	Raghav G Uddihal	Support defect fixes and triage
Product Owner	Prithvish Adhikari	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from dev team
Test environment downtime	Run the full stack locally with Docker Compose as backup
Payment gateway sandbox unavailable	Use a mock gateway with the same API and signed webhooks
Concurrency defects hard to reproduce	Automated parallel-request tests and database constraints

11. Assumptions & Dependencies

- Payment gateway sandbox and SMTP sandbox will be stable and available
- Test data (users, vehicles, lots, slots, tariffs) will be seeded before execution
- Requirements are frozen at SRS v1.0 for this test cycle

12. Suspension & Resumption Criteria

Suspend testing if:
- Environment unavailable for >4 hours
- Build is too unstable (blocks >30% test cases)

Resume testing if:
- Blocking defects are resolved
- Environment stabilized

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases.
Example:
- SPMS-F-001 (Driver registration) → TC-AUTH-01
- SPMS-F-002 (Login & JWT issue) → TC-AUTH-02
- SPMS-F-003 (Role-based access control) → TC-AUTH-03
- SPMS-F-004 (Account lockout after 5 failures) → TC-AUTH-04
- SPMS-F-010 (Manage lots / levels / slots) → TC-LOT-01
- SPMS-F-011 (Configure tariffs) → TC-LOT-02
- SPMS-F-012 (Slot maintenance status) → TC-LOT-03
- SPMS-F-020 (Display free-slot counts) → TC-AVL-01
- SPMS-F-021 (Live status push ≤ 5 s) → TC-AVL-02
- SPMS-F-030 (Reserve a slot) → TC-RES-01
- SPMS-F-031 (Prevent double booking) → TC-RES-02
- SPMS-F-032 (Modify / cancel with refund) → TC-RES-03
- SPMS-F-033 (Auto-release no-shows) → TC-RES-04
- SPMS-F-040 (Auto-allocate suitable slot) → TC-ALC-01
- SPMS-F-041 (Attendant allocation override) → TC-ALC-02
- SPMS-F-050 (Record vehicle entry) → TC-GATE-01
- SPMS-F-051 (Validate reservation QR) → TC-GATE-02
- SPMS-F-052 (Reject duplicate entry) → TC-GATE-03
- SPMS-F-053 (Record vehicle exit) → TC-GATE-04
- SPMS-F-060 (Fee calculation) → TC-PAY-01
- SPMS-F-061 (Online & cash payment) → TC-PAY-02
- SPMS-F-062 (Digital receipt) → TC-PAY-03
- SPMS-F-063 (Block unpaid exit / waiver) → TC-PAY-04
- SPMS-F-070 (Occupancy & revenue dashboard) → TC-RPT-01
- SPMS-F-071 (CSV export) → TC-RPT-02
- SPMS-F-072 (Email notifications) → TC-NOT-01
- SPMS-F-073 (Driver history & receipts) → TC-HIS-01
- SPMS-NF-001 (API response time) → TC-PERF-01
- SPMS-NF-002 (Gate transaction time) → TC-PERF-02
- SPMS-NF-003 (Availability & backup) → TC-REL-01
- SPMS-NF-004 (Usability & accessibility) → TC-UX-01

- SPMS-NF-005 (Scalability) → TC-PERF-03
- SPMS-NF-006 (Maintainability) → TC-MNT-01
- SPMS-SR-001 (TLS 1.2+ everywhere) → TC-SEC-01
- SPMS-SR-002 (bcrypt password hashing) → TC-SEC-02
- SPMS-SR-003 (JWT expiry & revocation) → TC-SEC-03
- SPMS-SR-004 (Input validation / no SQLi-XSS) → TC-SEC-04
- SPMS-SR-005 (No card data; signed webhooks) → TC-SEC-05
- SPMS-SR-006 (Rate limiting) → TC-SEC-06
- SPMS-SR-007 (Append-only audit log) → TC-SEC-07

14. Test Metrics & Reporting

Metrics collected:
- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:
- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Rahul Prasanna	Rahul/11-10-2026
Test Engineer	Rufus Thomas	Rufus/11-10-2026
Dev Lead	Raghav G Uddihal	Raghav/11-10-2026
Product Owner	Prithvish Adhikari	Prithvish/11-10-2026